

WOXSEN UNIVERSITY -- SCHOOL OF TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

DBMS LAB RECORD: COMPACT OUTPUT STRIPS (EXP 1 - 13)

Course Code: 25TU03MJM0 | 10-Mark Record Book Submission Pack

[NOTE] HOW TO PASTE 3 TO 5 OUTPUTS ON ONE RECORD PAGE:

- 1. Print this PDF on single-sided A4 paper.
- 2. Cut each experiment strip along the dashed boundary line [CUT].
- 3. Strips are made micro-sized (~35mm-50mm high) so 3 to 5 strips stack neatly vertically.
- 4. Experiment 5 includes ALL Joins (Venn Graph, Single/Multi Aliases, Inner/Left/Right Joins & Union).
- 5. Paste them vertically onto the blank left page facing each experiment in your record book.

EXPERIMENT STRIP CHECKLIST & PAGE GUIDE

Exp No	Experiment Title	Strips Count
EXP 01	Hotel Booking System Attribute Matrix	1 Strip
EXP 02	Basic SQL CRUD Operations (Table, SELECT, UPDATE)	3 Strips
EXP 03	More SQL Queries (DISTINCT, BETWEEN, ORDER BY, LIMIT)	3 Strips
EXP 04	Table Constraints & Aggregates (MIN, MAX, AVG, SUM, COUNT)	3 Strips
EXP 05	ALL Joins (Venn Graph, Aliases, Inner, Left/Right, Union)	5 Strips
EXP 06	LIKE Wildcards (% and _), GROUP BY & HAVING	3 Strips
EXP 07	SQL Views (HighEarning) & Stored Procedures (CALL AddNew)	2 Strips
EXP 08	Scenario-Based Practice (12 Tasks: LIKE, AVG, EXISTS, ALL)	3 Strips
EXP 09	MongoDB CRUD Operations (createCollection, find, updateOne)	3 Strips
EXP 10	MongoDB Query Operators (\$gt, \$or, \$regex)	3 Strips
EXP 11	MongoDB Update Operations (\$inc, \$currentDate, \$push)	2 Strips
EXP 12	MongoDB Aggregations (\$group sum, \$match & sort pipeline)	2 Strips
EXP 13	MongoDB Search and Indexing (Text Search, dropIndex)	2 Strips

[CUT] CUT HERE [CUT]

EXP 01 [1/1]: Hotel Booking System Attribute Matrix

Query/Cmd: Entity Identification & Normalization

```
+-----+-----+
| Entity Name | Identified Attributes & Constraints |
+-----+-----+
| GUEST       | GuestID (PK), FirstName, LastName, Phone, Email, Address, IdProof |
| ROOM        | RoomID (PK), RoomNumber (UNIQUE), RoomType (Single/Double/Suite), Rate, Status |
| RESERVATION | ReservationID (PK), GuestID (FK), RoomID (FK), CheckInDate, CheckOutDate |
| PAYMENT     | PaymentID (PK), ReservationID (FK), PaymentDate, Amount, Method (Card/UPI) |
+-----+-----+
```

[STICK] WHERE TO STICK: Paste on LEFT (BLANK) PAGE facing Experiment 1 Aim.

[CUT] CUT HERE [CUT]

EXP 02 [1/3]: Basic CRUD Operations -- 10 Records Inserted

Query/Cmd: SELECT * FROM Students;

```
+-----+-----+-----+-----+-----+-----+
| Name   | ID   | Course | DOB    | Gender | BloodGroup |
+-----+-----+-----+-----+-----+-----+
| Abhiram | 1001 | CSE-AIML | 101003 | M      | O+ve      |
| Abhimanyu | 1002 | CSE      | 010303 | M      | B+ve      |
| Rudra   | 1003 | CSE-DS   | 210503 | M      | O+ve      |
| Nandani | 1004 | CSE      | 280303 | F      | B+ve      |
| Ridhima | 1005 | CSE-BIC  | 031003 | F      | O+ve      |
+-----+-----+-----+-----+-----+-----+ (10 rows in set)
```

[STICK] WHERE TO STICK: Paste on LEFT (BLANK) PAGE facing Exp 2 (Top - Strip 1 of 3).

[CUT] CUT HERE [CUT]

EXP 02 [2/3]: Conditional Selection with WHERE Clause

Query/Cmd: SELECT * FROM Students WHERE ID = 1005;

```
+-----+-----+-----+-----+-----+-----+
| Name   | ID   | Course | DOB    | Gender | BloodGroup |
+-----+-----+-----+-----+-----+-----+
| Ridhima | 1005 | CSE-BIC | 031003 | F      | O+ve      |
+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

[STICK] WHERE TO STICK: Paste on LEFT (BLANK) PAGE facing Exp 2 (Middle - Strip 2 of 3).

[CUT] CUT HERE [CUT]

EXP 02 [3/3]: Record Update & Verification

Query/Cmd: UPDATE Students SET Course='CSE', BloodGroup='B+ve' WHERE ID=1005;

Query OK, 1 row affected (0.01 sec) Rows matched: 1 Changed: 1
mysql> SELECT * FROM Students WHERE ID = 1005; -- [AFTER UPDATE]

```
+-----+-----+-----+-----+-----+-----+
| Name   | ID   | Course | DOB    | Gender | BloodGroup |
+-----+-----+-----+-----+-----+-----+
| Ridhima | 1005 | CSE     | 031003 | F      | B+ve      |
+-----+-----+-----+-----+-----+-----+
```

[STICK] WHERE TO STICK: Paste on LEFT (BLANK) PAGE facing Exp 2 (Bottom - Strip 3 of 3).

[CUT] CUT HERE [CUT]

EXP 03 [1/3]: DISTINCT Clause Output

Query/Cmd: SELECT DISTINCT Course Name FROM Student;

```
+-----+
| Course_Name |
+-----+
| CSE          |
| CSE-AIML     |
| CSE-DS       |
| CSE-BIC      |
+-----+
4 rows in set (0.00 sec) -- Duplicates eliminated
```

[STICK] WHERE TO STICK: Paste on LEFT (BLANK) PAGE facing Exp 3 (Top - Strip 1 of 3).

[CUT] CUT HERE [CUT]

EXP 03 [2/3]: WHERE Clause with BETWEEN Operator

Query/Cmd: SELECT * FROM Student WHERE sID BETWEEN 1002 AND 1004;

```
+-----+-----+-----+-----+-----+
| sID | sName | Course_Name | Address | email |
+-----+-----+-----+-----+-----+
| 1002 | Raj | CSE-AIML | Hyderabad | NULL |
| 1003 | Ajay | CSE-DS | Sadashivapet | NULL |
| 1004 | Sandhya | CSE-BIC | Kurnool | NULL |
+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

[STICK] WHERE TO STICK: Paste on LEFT (BLANK) PAGE facing Exp 3 (Middle - Strip 2 of 3).

[CUT] CUT HERE [CUT]

EXP 03 [3/3]: ORDER BY DESC & LIMIT 3 (Select TOP)

Query/Cmd: SELECT * FROM Student ORDER BY sID DESC LIMIT 3;

```
+-----+-----+-----+-----+-----+
| sID | sName | Course_Name | Address | email |
+-----+-----+-----+-----+-----+
| 1005 | Ruhanika | CSE-BIC | Nandyal | NULL |
| 1004 | Sandhya | CSE-BIC | Kurnool | NULL |
| 1003 | Ajay | CSE-DS | Sadashivapet | NULL |
+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

[STICK] WHERE TO STICK: Paste on LEFT (BLANK) PAGE facing Exp 3 (Bottom - Strip 3 of 3).

[CUT] CUT HERE [CUT]

EXP 04 [1/3]: Table Constraints & Filter (Employee)

Query/Cmd: SELECT emp no, emp name, job, salary FROM Employee WHERE job IN ('Manager', 'Clerk');

```
+-----+-----+-----+-----+-----+
| emp_no | emp_name | job | salary | dept_no |
+-----+-----+-----+-----+-----+
| 201001 | Anand | Manager | 50000 | 1 |
| 201002 | Vamsi | Manager | 50000 | 2 |
| 201003 | Sanjana | Clerk | 20000 | 4 |
| 201004 | Nanditha | Clerk | 20000 | 4 |
| 221001 | Jaya | Manager | 50000 | 1 |
| 221005 | Sita | Manager | 50000 | 2 |
+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)
```

[STICK] WHERE TO STICK: Paste on LEFT (BLANK) PAGE facing Exp 4 (Top - Strip 1 of 3).

[CUT] CUT HERE [CUT]

EXP 04 [2/3]: Aggregate Functions (MIN, MAX, AVG, SUM, COUNT)

Query/Cmd: SELECT MIN(salary), MAX(salary), AVG(salary), SUM(salary), COUNT(*) FROM Employee;

```
+-----+-----+-----+-----+-----+
| min_sal | max_sal | avg_sal | total_payroll | total_employees |
+-----+-----+-----+-----+-----+
| 20000 | 50000 | 36333.33 | 545000 | 15 |
+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

[STICK] WHERE TO STICK: Paste on LEFT (BLANK) PAGE facing Exp 4 (Middle - Strip 2 of 3).

[CUT] CUT HERE [CUT]

EXP 04 [3/3]: Top Salary & String Concatenation

Query/Cmd: SELECT emp name, salary FROM Employee ORDER BY salary DESC LIMIT 1;

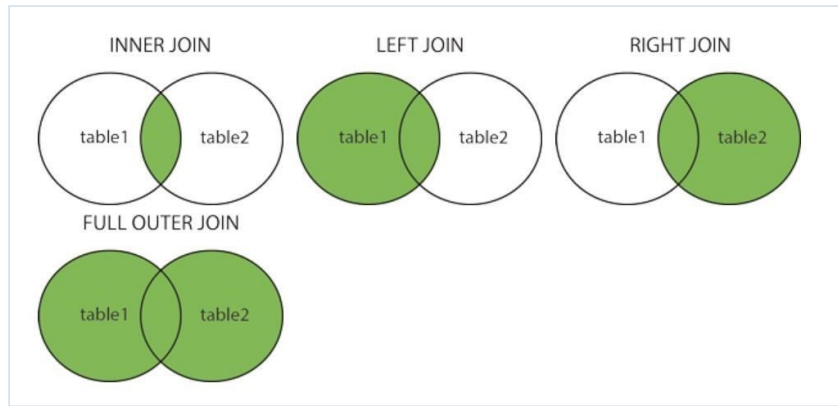
```
+-----+-----+
| emp_name | salary |
+-----+-----+
| Anand | 50000 |
+-----+-----+
1 row in set (0.00 sec) (Top earner matching MAX constraint)
```

[STICK] WHERE TO STICK: Paste on LEFT (BLANK) PAGE facing Exp 4 (Bottom - Strip 3 of 3).

[CUT] CUT HERE [CUT]

EXP 05 [GRAPH 1/5]: SQL JOINS GRAPHICAL VENN DIAGRAM

INNER JOIN | LEFT JOIN | RIGHT JOIN | FULL OUTER JOIN



[STICK] WHERE TO STICK: Paste at TOP of LEFT (BLANK) PAGE facing Exp 5 (Strip 1 of 5).

[CUT] CUT HERE [CUT]

EXP 05 [2/5]: SQL Aliases ('AS' Keyword & Bracketed Space)

Query/Cmd: SELECT Name AS Student Name...; SELECT ID AS Student ID, BloodGroup AS [Blood Group]...

```
mysql> SELECT Name AS Student_Name FROM Students LIMIT 3;
+-----+
| Student_Name |
+-----+
| Abhiram      |
| Abhimanyu    |
| Rudra        |
+-----+

mysql> SELECT ID AS Student_ID, BloodGroup AS [Blood Group] FROM Students LIMIT 2;
+-----+-----+
| Student_ID | Blood Group |
+-----+-----+
| 1001       | O+ve       |
| 1002       | B+ve       |
+-----+-----+
```

[STICK] WHERE TO STICK: Paste on LEFT (BLANK) PAGE facing Exp 5 (Strip 2 of 5 - Aliases).

[CUT] CUT HERE [CUT]

EXP 05 [3/5]: INNER JOIN (Orders & Customers)

Query/Cmd: SELECT Orders.OrderID, Customers.CustomerName FROM Orders INNER JOIN Customers...

```
+-----+-----+-----+
| OrderID | CustomerName | OrderDate |
+-----+-----+-----+
| 10308   | Ana Trujillo | 2023-09-18 |
| 10309   | Antonio Moreno | 2023-09-19 |
| 10310   | Around the Horn | 2023-09-20 |
| 10311   | Berglunds snabbkop | 2023-09-21 |
+-----+-----+-----+

4 rows in set (0.00 sec) -- Only matching records from both tables
```

[STICK] WHERE TO STICK: Paste on LEFT (BLANK) PAGE facing Exp 5 (Strip 3 of 5 - Inner Join).

[CUT] CUT HERE [CUT]

EXP 05 [4/5]: LEFT & RIGHT OUTER JOIN (Preserving NULLs)

Query/Cmd: SELECT Customers.CustomerName, Orders.OrderID FROM Customers LEFT JOIN Orders...

```
+-----+-----+
| CustomerName | OrderID |
+-----+-----+
| Alfreds Futterkiste | NULL | <-- Customer with 0 orders
| Ana Trujillo | 10308 |
| Antonio Moreno | 10309 |
| Around the Horn | 10310 |
+-----+-----+

4 rows in set (0.00 sec) -- Preserves unmatched rows with NULL
```

[STICK] WHERE TO STICK: Paste on LEFT (BLANK) PAGE facing Exp 5 (Strip 4 of 5 - Outer Joins).

[CUT] CUT HERE [CUT]

EXP 05 [5/5]: UNION vs UNION ALL (Distinct vs All Rows)

Query/Cmd: SELECT Course FROM Students UNION [ALL] SELECT Course Name FROM Student;

```
mysql> -- [UNION: Distinct 4 rows]      mysql> -- [UNION ALL: 15 rows with duplicates]
+-----+                                +-----+
| Course |                                | Course | (CSE-AIML, CSE, CSE-DS, CSE,
+-----+                                +-----+ CSE-BIC, CSE-DS, CSE-BIC...)
| CSE-AIML |                                15 rows in set (0.00 sec)
| CSE      |
| CSE-DS   |
| CSE-BIC  |
+-----+ (4 rows in set)
```

[STICK] WHERE TO STICK: Paste at BOTTOM of LEFT (BLANK) PAGE facing Exp 5 (Strip 5 of 5).

[CUT] CUT HERE [CUT]

EXP 06 [1/3]: LIKE Pattern Search (% and _ Wildcards)

Query/Cmd: SELECT Name, Course FROM Students WHERE Course LIKE '%AIML%';

```
+-----+-----+
| Name      | Course  |
+-----+-----+
| Abhiram   | CSE-AIML | (Matches 'Ab%' and '%AIML%')
| Abhimanyu | CSE      | (Matches 'Ab%')
| Veer      | CSE-AIML | (Matches '%AIML%')
+-----+-----+
```

Pattern search with % (any chars) and _ (single char) verified

[STICK] WHERE TO STICK: Paste on LEFT (BLANK) PAGE facing Exp 6 (Top - Strip 1 of 3).

[CUT] CUT HERE [CUT]

EXP 06 [2/3]: GROUP BY Job & Headcount

Query/Cmd: SELECT job, COUNT(emp no) AS count_emp FROM Employee GROUP BY job;

```
+-----+-----+
| job      | count_emp |
+-----+-----+
| Manager  | 4         |
| Assistant| 3         |
| Engineer | 3         |
| Clerk    | 2         |
| Analyst  | 2         |
+-----+-----+
```

[STICK] WHERE TO STICK: Paste on LEFT (BLANK) PAGE facing Exp 6 (Middle - Strip 2 of 3).

[CUT] CUT HERE [CUT]

EXP 06 [3/3]: HAVING Group Filter & ORDER BY

Query/Cmd: SELECT job, COUNT(emp no) AS total, AVG(salary) GROUP BY job HAVING total >= 2;

```
+-----+-----+
| job      | total | avg_salary |
+-----+-----+
| Manager  | 4     | 50000.00  |
| Assistant| 3     | 25000.00  |
| Engineer | 3     | 40000.00  |
| Clerk    | 2     | 20000.00  |
| Analyst  | 2     | 35000.00  |
+-----+-----+
```

5 rows in set (0.00 sec) -- Filtered out job roles where count < 2

[STICK] WHERE TO STICK: Paste on LEFT (BLANK) PAGE facing Exp 6 (Bottom - Strip 3 of 3).

[CUT] CUT HERE [CUT]

EXP 07 [1/2]: Database VIEWS Execution

Query/Cmd: SELECT * FROM HighEarningEmployees; SELECT * FROM DepartmentSalaryExpenses;

```
+-----+-----+-----+-----+-----+-----+
| EmployeeID | Name           | Salary | DepartmentName |
+-----+-----+-----+-----+-----+-----+
| 4          | Michael Brown | 7000   | IT              |
| 5          | Sarah Johnson | 7500   | Marketing       |
+-----+-----+-----+-----+-----+-----+
| DepartmentName | TotalExpense |
+-----+-----+-----+-----+-----+-----+
| Sales          | 15700        |
| Marketing      | 13500        |
| IT             | 7000         |
+-----+-----+-----+-----+-----+-----+
```

[STICK] WHERE TO STICK: Paste on LEFT (BLANK) PAGE facing Exp 7 (Top - Strip 1 of 2 - Views).

[CUT] CUT HERE [CUT]

EXP 07 [2/2]: Stored Procedure Execution (CALL AddNewEmployee)

Query/Cmd: CALL AddNewEmployee('Anna Taylor', 103, 6800);

Query OK, 1 row affected (0.01 sec)

mysql> SELECT EmployeeID, Name, DepartmentID, Salary FROM Employees WHERE Name = 'Anna Taylor';

```
+-----+-----+-----+-----+
| EmployeeID | Name           | DepartmentID | Salary |
+-----+-----+-----+-----+
| 7          | Anna Taylor    | 103          | 6800   |
+-----+-----+-----+-----+
```

1 row in set (0.00 sec) -- Stored procedure executed successfully

[STICK] WHERE TO STICK: Paste on LEFT (BLANK) PAGE facing Exp 7 (Bottom - Strip 2 of 2 - Procedures).

[CUT] CUT HERE [CUT]

EXP 08 [1/3]: Name Filter ('%e') & Department Statistics

Query/Cmd: SELECT Name WHERE Name LIKE '%e'; SELECT Department, COUNT(*), AVG(Salary)...

```
+-----+
| Name |
+-----+
| Alice |
| Eve |
| Grace |
+-----+

+-----+-----+-----+-----+
| Department | EmployeeCount | AvgSalary |
+-----+-----+-----+-----+
| IT | 3 | 67333.33 |
| HR | 2 | 51500.00 |
| Marketing | 2 | 72500.00 |
+-----+-----+-----+-----+
```

[STICK] WHERE TO STICK: Paste on LEFT (BLANK) PAGE facing Exp 8 (Top - Strip 1 of 3).

[CUT] CUT HERE [CUT]

EXP 08 [2/3]: Correlated Subquery with EXISTS

Query/Cmd: SELECT Name FROM Employees e WHERE EXISTS (SELECT 1 FROM Projects p WHERE p.Budget > 50000);

```
+-----+
| Name |
+-----+
| Frank |
| Grace |
+-----+

2 rows in set (0.00 sec) -- Employees on projects with budget > $50,000
```

[STICK] WHERE TO STICK: Paste on LEFT (BLANK) PAGE facing Exp 8 (Middle - Strip 2 of 3).

[CUT] CUT HERE [CUT]

EXP 08 [3/3]: Set Comparison Subquery (Salary > ALL in HR)

Query/Cmd: SELECT Name FROM Employees WHERE Salary > ALL (SELECT Salary FROM Employees WHERE Department='HR');

```
+-----+-----+-----+-----+
| Alice | Bob | Eve | Frank | Grace |
+-----+-----+-----+-----+

5 rows in set (0.00 sec) -- All employees with Salary > $55,000
```

[STICK] WHERE TO STICK: Paste on LEFT (BLANK) PAGE facing Exp 8 (Bottom - Strip 3 of 3).

[CUT] CUT HERE [CUT]

EXP 09 [1/3]: MongoDB Collection Creation & insertMany

Query/Cmd: db.createCollection('Lab9'); db.Lab9.insertMany([...]);

```
CollegeDB> db.createCollection('Lab9')
{ ok: 1 }
CollegeDB> db.Lab9.insertMany([ { id: 1001, name: 'Abhiram', course: 'CSE-AI ML', marks: 88 }, ... ])
{ acknowledged: true, insertedIds: [ ObjectId('672c4101a88246916737a101'), ... ] }
```

[STICK] WHERE TO STICK: Paste on LEFT (BLANK) PAGE facing Exp 9 (Top - Strip 1 of 3).

[CUT] CUT HERE [CUT]

EXP 09 [2/3]: MongoDB find() Document Retrieval

Query/Cmd: db.Lab9.find({ id: 1005 });

```
[ { _id: ObjectId('...'), id: 1005, name: 'Ridhima', course: 'CSE-BIC', marks: 92 } ]
```

[STICK] WHERE TO STICK: Paste on LEFT (BLANK) PAGE facing Exp 9 (Middle - Strip 2 of 3).

[CUT] CUT HERE [CUT]

EXP 09 [3/3]: MongoDB updateOne & deleteOne Operations

Query/Cmd: db.Lab9.updateOne({ id: 1005 }, { \$set: { marks: 96 } });

```
{ acknowledged: true, matchedCount: 1, modifiedCount: 1 }
Document with id: 1005 updated with marks: 96
```

[STICK] WHERE TO STICK: Paste on LEFT (BLANK) PAGE facing Exp 9 (Bottom - Strip 3 of 3).

[CUT] CUT HERE [CUT]

EXP 10 [1/3]: Comparison Operators (\$gt & \$gte)

Query/Cmd: db.Lab9.find({ marks: { \$gt: 85 } }, { _id: 0, id: 1, name: 1, marks: 1 });

```
[
  { id: 1001, name: 'Abhiram', marks: 88 },
  { id: 1003, name: 'Rudra', marks: 91 },
  { id: 1005, name: 'Ridhima', marks: 96 },
  { id: 1008, name: 'Kshithija', marks: 89 }
]
```

[STICK] WHERE TO STICK: Paste on LEFT (BLANK) PAGE facing Exp 10 (Top - Strip 1 of 3).

[CUT] CUT HERE [CUT]

EXP 10 [2/3]: Logical Disjunction (\$or Operator)

Query/Cmd: db.Lab9.find({ \$or: [{ course: 'CSE-AIML' }, { marks: { \$gte: 95 } }] });

```
[
  { id: 1001, name: 'Abhiram', course: 'CSE-AIML' },
  { id: 1005, name: 'Ridhima', course: 'CSE-BIC' },
  { id: 1009, name: 'Veer', course: 'CSE-AIML' }
]
```

[STICK] WHERE TO STICK: Paste on LEFT (BLANK) PAGE facing Exp 10 (Middle - Strip 2 of 3).

[CUT] CUT HERE [CUT]

EXP 10 [3/3]: Regex Pattern Search (\$regex: /^A/i)

Query/Cmd: db.Lab9.find({ name: { \$regex: /^A/i } });

```
[ { id: 1001, name: 'Abhiram' }, { id: 1002, name: 'Abhimanyu' } ]
2 documents matched regex condition /^A/i
```

[STICK] WHERE TO STICK: Paste on LEFT (BLANK) PAGE facing Exp 10 (Bottom - Strip 3 of 3).

[CUT] CUT HERE [CUT]

EXP 11 [1/2]: Field Update Modifiers (\$inc, \$currentDate)

Query/Cmd: db.sampledata.updateOne({ name: 'Emily Davis' }, { \$inc: { age: 1, salary: 5000 }, \$currentDate... });

```
{ acknowledged: true, matchedCount: 1, modifiedCount: 1 }
sampleDB> db.sampledata.find({ name: 'Emily Davis' })
[ { name: 'Emily Davis', age: 30, salary: 75000, lastUpdated: ISODate('2026-10-02T08:00:00Z') } ]
```

[STICK] WHERE TO STICK: Paste on LEFT (BLANK) PAGE facing Exp 11 (Top - Strip 1 of 2).

[CUT] CUT HERE [CUT]

EXP 11 [2/2]: Array Modifiers (\$push & \$pull)

Query/Cmd: db.sampledata.updateOne({ name: 'Emily Davis' }, { \$push: { skills: 'MongoDB' } });

```
{ acknowledged: true, matchedCount: 1, modifiedCount: 1 }
Resulting Skills Array: [ 'Python', 'SQL', 'MongoDB' ]
```

[STICK] WHERE TO STICK: Paste on LEFT (BLANK) PAGE facing Exp 11 (Bottom - Strip 2 of 2).

[CUT] CUT HERE [CUT]

EXP 12 [1/2]: Aggregation Pipeline (\$group & \$sum)

Query/Cmd: db.sales.aggregate([{\$group: { _id: '\$item', totalQty: { \$sum: '\$quantity' }, totalRevenue... } }]);

```
[
  { _id: 'apple', totalQty: 80, totalRevenue: 96.0 },
  { _id: 'banana', totalQty: 150, totalRevenue: 120.0 },
  { _id: 'orange', totalQty: 75, totalRevenue: 112.5 }
]
```

[STICK] WHERE TO STICK: Paste on LEFT (BLANK) PAGE facing Exp 12 (Top - Strip 1 of 2).

[CUT] CUT HERE [CUT]

EXP 12 [2/2]: Filtered Pipeline (\$match, \$sort, \$limit)

Query/Cmd: db.sales.aggregate([{\$match: { status: 'A' } }, {\$sort: { quantity: -1 } }, {\$limit: 2 }]);

```
[
  { item: 'banana', price: 0.8, quantity: 100, status: 'A' },
  { item: 'apple', price: 1.2, quantity: 50, status: 'A' }
]
```

[STICK] WHERE TO STICK: Paste on LEFT (BLANK) PAGE facing Exp 12 (Bottom - Strip 2 of 2).

[CUT] CUT HERE [CUT]

EXP 13 [1/2]: Full-Text Indexing & \$text Search

Query/Cmd: db.articles.createIndex({ content: 'text' }); db.articles.find({ \$text: { \$search: 'database' } });

```
companyDB> db.articles.createIndex({ content: 'text' })
{ createdCollectionAutomatically: false, numIndexesBefore: 1, numIndexesAfter: 2, ok: 1 }
companyDB> db.articles.find({ $text: { $search: 'database' } })
[
  { content: 'MongoDB is a NoSQL database.' },
  { content: 'Databases like MongoDB are popular for modern web applications.' }
]
```

[STICK] WHERE TO STICK: Paste on LEFT (BLANK) PAGE facing Exp 13 (Top - Strip 1 of 2).

[CUT] CUT HERE [CUT]

EXP 13 [2/2]: Compound Index Creation & dropIndex

Query/Cmd: db.employees.createIndex({ last name: 1, date of birth: 1 }); db.employees.dropIndex('first name 1');

```
companyDB> db.employees.createIndex({ last_name: 1, date_of_birth: 1 })
{ numIndexesBefore: 2, numIndexesAfter: 3, ok: 1 }
companyDB> db.employees.dropIndex('first_name_1')
{ nIndexesWas: 3, ok: 1 }
```

[STICK] WHERE TO STICK: Paste on LEFT (BLANK) PAGE facing Exp 13 (Bottom - Strip 2 of 2).